

Lights off only when the room is really empty.

EcoSwitch fuses a cheap PIR motion sensor with a 24 GHz mmWave radar so a person sitting still is never plunged into darkness, and empty classrooms stop wasting power. No cameras. Works offline.

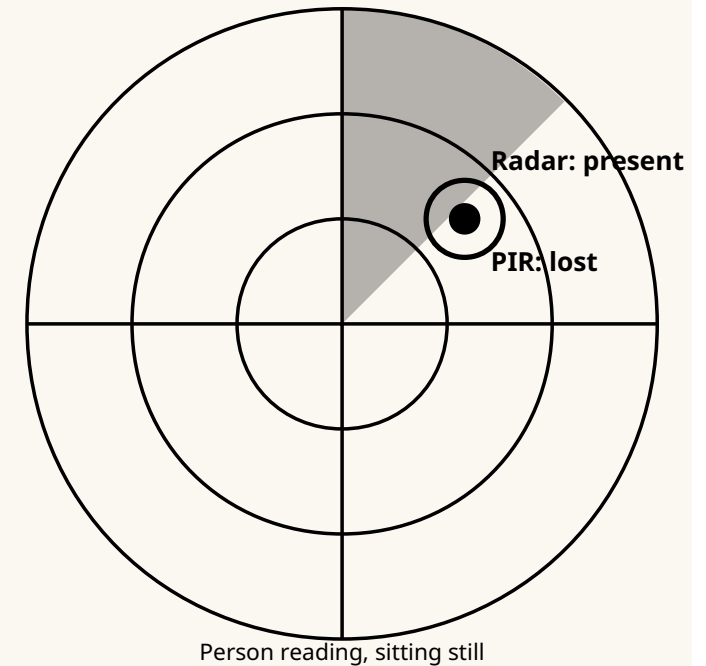
B.Tech Year 1, IT

PCCOE, Pune

Guide: Rohit Tathe

~12 weeks

5-member team



Abstract

Lights and fans in classrooms, labs and offices are often left running in empty rooms. PIR sensors are cheap but respond to movement, so they can switch off on someone reading or writing an exam. Cameras fix that but cost more and raise privacy concerns.

EcoSwitch is a low-cost, offline controller that turns loads off only after **both** sensors agree the room has been empty for a set time. The team builds a prototype with a fan and lights (low-voltage for safety), then tests hybrid vs PIR-only vs mmWave-only on false-vacancy rate, false-occupancy rate, response time and energy saved.

2,000 kWh/yr

Low case waste, 60 rooms. Does not pay back.

9,100 kWh/yr

Base case waste. Payback ~3.5 years.

27,000 kWh/yr

High case waste. Payback under 1 year.

The value depends on how much waste really occurs, which is why a measurement audit (Phase 0) comes first.

Background and problem

Buildings use a large share of electricity. Schools run on fixed timetables with gaps (between lectures, breaks, after practicals, early and late hours) where lights and fans stay on because switching off is nobody's explicit job to be validated by audit.

Surveys of occupancy sensing (Nguyen & Aiello 2013; Dong et al. 2019; Chen et al. 2018) cover cost, accuracy, privacy and installation trade-offs. A lighting-control meta-analysis (Williams et al. 2012) found roughly a quarter average lighting savings, mostly in commercial buildings abroad, so it cannot be applied directly to an Indian campus.

Problem: manual or timer switching wastes energy; motion-only automation makes false "vacant" calls on still occupants, annoying users until they bypass or remove it. We need a low-cost, privacy-preserving approach that stays reliable for stationary people.

Energy loss calculator

Annual waste = rooms × load (kW) × vacant-but-ON hours/day × 220 days. Pick a scenario or drag the sliders. Each node is assumed to draw 1.5 W continuously. Tariff ₹8/kWh, deployment ₹3,000/room, 0.75 kg CO₂/kWh. These are assumptions to verify, not measurements.

9,108 kWh

Base-case gross annual waste (illustrative)

₹72,864

Base-case gross annual cost (illustrative)

6,498 kWh

Base-case net annual savings after node use

3.5 yrs

Illustrative simple payback

Table 1: Scenario assumptions

Parameter	Low	Base	High
Lights + fans per room	100 W lights + 200 W fans = 300 W	8 × 20 W LED + 4 × 75 W fans = 460 W	~280 W older tubes + 5 × 80 W fans = 680 W
Vacant-but-ON h/room/day	0.5 h	1.5 h	3 h
Share of waste captured	60%	80%	90%

Table 2: Estimated annual loss (lights and fans)

	Low	Base	High
Waste per room per day	0.15 kWh	0.69 kWh	2.04 kWh
Waste per room per year	33 kWh	151.8 kWh	448.8 kWh
Gross waste, 60 rooms	1,980 kWh	9,108 kWh	26,928 kWh
Gross cost at ₹8/kWh	₹15,840	₹72,864	₹2,15,424

Table 3: Recoverable savings, net of node consumption

	Low	Base	High
Recovered waste	1,188 kWh	7,286 kWh	24,235 kWh
Minus nodes' own use	–788 kWh	–788 kWh	–788 kWh
Net savings	~400 kWh	~6,498 kWh	~23,447 kWh
Net savings per year	~₹3,200	~₹51,984	~₹1,87,574
Deployment (60 × ₹3,000)	₹1,80,000	₹1,80,000	₹1,80,000
Simple payback	~56 years	~3.5 years	~1 year
CO ₂ avoided (0.75 kg/kWh)	~0.3 t	~4.9 t	~17.6 t

Takeaways. Savings scale with vacant-but-ON hours; in a well-run campus (Low) it is not worth installing. Standby use (~788 kWh/yr) is about 11% of base-case recovered energy, so low-power design matters. AC would add about 2,640 kWh and ₹21,000/yr for 10 rooms (1.5 kW, 1 h/day, 80% capture) but needs compressor-safe handling, so it is out of scope.

Existing solutions

PIR switches

Widely sold wall and ceiling occupancy switches.

Dual-technology

PIR plus ultrasonic or microphonic sensing.

BMS

Scheduling, CO₂ or camera-based occupancy.

ECBC

India's building code has lighting-control provisions; read the clauses for educational buildings.

Why they are not more common (hypotheses to test with the college electrician)

- Tariffs and budgets make payback unattractive unless waste is high.
- Retrofits need wiring changes and trained labour.
- PIR-only systems switch off on seated people, so users bypass them.
- Split incentive: whoever pays the bill is not who chooses or uses the equipment.
- Cameras raise privacy concerns and need more processing.

Sensing technologies

Table 4: relevance ratings are the team's judgement for this application. Filter by minimum stars.

Technology	Detects	Strength	Caveat	Relevance
PIR	Infrared change / movement	Very cheap, simple, low power	Can miss a stationary person	★ ★ ★ ★
Ultrasonic	Distance to reflector	Cheap, simple ranging	Cannot identify a person; furniture interferes	★ ★ ★
IR beam-break	Beam interruption	Deterministic at fixed points	Needs alignment; only interruption	★ ★ ★ ★
mmWave radar	Reflections: motion and presence	Detects relatively stationary people	Placement and calibration matter	★ ★ ★ ★ ★
Camera	Visual presence, count, location	Rich information	Privacy, lighting, processing	★ ★
Thermal camera	Heat signatures	Works without light	Costly; privacy concerns	★ ★
CO ₂ sensor	Air CO ₂ change	Aggregate occupancy	Slow; ventilation affects it	★ ★
Smart meter / current	Electrical consumption	Real load data	Consumption is not occupancy	★ ★ ★
Wi-Fi / BLE	Device presence	Reuses infrastructure	People may carry no device	★ ★

Cameras and thermal cameras lose on privacy, cost and processing. CO₂ is too slow. Wi-Fi/ BLE misses people without devices. Beam-break needs alignment and counting. PIR and mmWave balance cost, privacy and capability, and their weaknesses differ, which suggests combining them.

The hybrid solution

Situation	PIR	mmWave radar	Hybrid
Person walks in	Detects quickly	Detects	Detected quickly
Person sits still	May lose detection	Detects breathing and posture shifts (with limits)	Stays ON
Empty room, fan or curtain moving	Usually unaffected	May false-detect if untuned	Needs tuning
Sensor failure	Reduced coverage	Reduced reliability	Degrades to one sensor
Cost and power	Very low	Moderate	Moderate

PIR reacts to infrared change from moving warm bodies. 24 GHz FMCW radar modules like the HLK-LD2410 report moving and stationary targets (via micro-motion) with configurable range and sensitivity [from datasheets, to confirm](#) .

Turn ON

As soon as either sensor sees presence, so nobody is left in the dark.

Stay ON

While either sensor still reports presence.

Turn OFF

Only after both sensors report none continuously for the timeout.

Try the controller

Switch the mode, then simulate a person who sits still: PIR drops out, the radar does not. Grace timeout here is 8 s (prototype: 20–30 s).

VACANT

Objectives and scope

Objectives

- Estimate waste with a transparent, measured-where-possible model.
- Build a prototype controlling a fan and lights from hybrid detection.
- Compare PIR-only, mmWave-only and hybrid.
- Measure false vacancy, false occupancy, response time, energy saved.
- Document for a larger deployment.

In scope

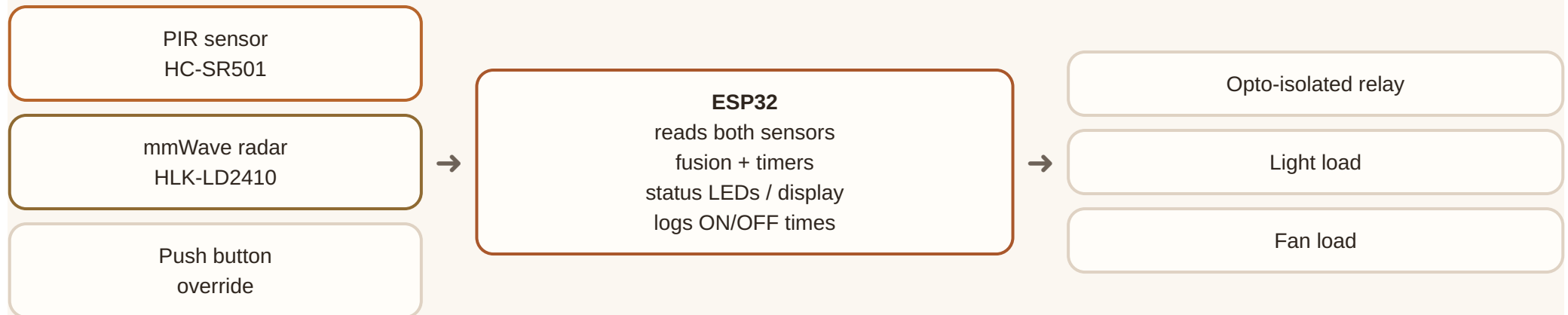
- One room or bench
- Two channels: lights, fan
- Offline local decisions
- Manual override
- Status display or logging
- Scenario tests, energy calculation

Out of scope

- Switching building mains (unless qualified supervision)
- AC integration
- People counting
- Cloud dashboards
- Machine learning
- Whole-institute install

System design

Architecture



Workflow: read sensors → classify OCCUPIED, GRACE or VACANT → switch relays → log events for energy and accuracy analysis.

Components (approx. ₹2,500–4,000 total, check current prices)

Component	Part	Role	Cost
Microcontroller	ESP32 dev board	Reads sensors, runs logic	₹500–700
Motion sensor	HC-SR501 PIR	Fast motion detection	₹80–150
Presence radar	HLK-LD2410 (24 GHz)	Moving and stationary presence	₹700–1,200
Switching	2-channel opto-isolated relay, 5 V	Switches fan and light	₹150–250
Demo loads	12 V DC fan, 12 V LED	Safe demo loads	₹300–500
Power	5 V and 12 V supplies	Powers system	₹300–500
Interface	Button, LEDs, optional OLED	Override and feedback	₹100–300
Misc.	Flyback diode, wiring, enclosure	Protection and assembly	₹300–500

Table 6: Pin connections (confirm on chosen board)

Signal	From	To	Note
PIR output	PIR OUT	GPIO 27	Set minimum module delay; use firmware timers
Radar UART	LD2410 TX	GPIO 16 (RX2)	Moving/stationary status, energy values
Radar UART	LD2410 RX	GPIO 17 (TX2)	Configuration
Radar out (optional)	LD2410 OUT	GPIO 26	Simple presence flag, backup
Lights relay	GPIO 23	Relay IN1	Check if active-low
Fan relay	GPIO 22	Relay IN2	
Manual override	Push button	GPIO 4	Input with pull-up
Status LED	GPIO 2	LED + resistor	

- Power the LD2410 and relay coil from stable 5 V with common ground; check the radar's logic levels.
- Put a flyback diode across the DC fan to protect relay contacts.
- Relay contacts switch the load side only.

Table 7: Controller states

State	Meaning	Entry	Loads	Exit
OCCUPIED	Someone present	PIR motion or radar presence	Light ON, fan ON	Both clear → GRACE
GRACE	Possibly vacant; countdown	Both sensors clear	Stay ON	Presence → OCCUPIED; timer ends → VACANT
VACANT	Room empty	Grace timer expired	Light OFF, fan OFF	PIR or radar presence → OCCUPIED
MANUAL OVERRIDE	User forced ON/OFF	Button pressed	As chosen	Timeout or button press
DEGRADED	One sensor not responding	Sensor fault or timeout	Working sensor with longer timeout; ON if both fail	Sensor recovers

Tunable parameters

Parameter	Prototype	Real room
Grace (off-delay) timeout	20–30 s	Likely several minutes
Radar sensitivity per gate	Tuned in testing	Depends on room and furniture
Radar range	Limited to test area	Matched to room size
Separate light/fan delays	Same initially	Could differ

Known engineering challenges

- **Fans and curtains** reflect radar: tune per-gate sensitivity, limit range, mount away from fan line of sight, test "empty room, fan running".
- **Placement** (height, angle) changes what sensors see: test several.
- **Sunlight and heat** can trigger PIR: test and document.
- **Through-wall or corridor detection** causes false occupancy: limit range and gates.

Why these technologies

Choice	Why	Alternatives
ESP32	Low cost, documented, enough GPIO/UART, Wi-Fi optional	Arduino Uno, Raspberry Pi
PIR HC-SR501	Cheapest, lowest-power motion detection	Ultrasonic, IR beam-break
mmWave LD2410	Low-cost stationary presence, no images	Camera, thermal, CO ₂
Opto-isolated relay	Isolates control side	Solid-state relay, transistor-only
Arduino framework (C++)	Easy for beginners	ESP-IDF, MicroPython
Offline decisions	No dependence on internet	Cloud control

Literature and research gap

Overviews

Nguyen & Aiello (2013), Dong et al. (2019), Chen et al. (2018): sensor range, accuracy limits, privacy and cost trade-offs.

In practice

Agarwal et al. (2010): PIR and door sensors drove building control in a university, so campus-scale control is feasible.

PIR limits and fusion

Labeodan et al. (2015) show PIR struggles with low motion; Dodier et al. (2006) show fusing sensors improves detection.

Savings potential

Williams et al. (2012): meaningful occupancy-sensor savings, US-focused, hence our scenarios plus local audit.

Radar

Hi-Link and TI documentation describe FMCW presence sensing. Add 2–3 peer-reviewed radar-occupancy papers.

Gap: a low-cost, camera-free, offline PIR + 24 GHz radar controller evaluated in an Indian classroom-like setting against PIR-only and radar-only baselines, with ceiling-fan interference in focus. Novelty is in implementation and evaluation, not the technologies.

Comparison with existing approaches

Criterion	Timer / manual	PIR-only	Camera	Cloud BMS	EcoSwitch
Cost	Very low	Low	High	High	Low–moderate
Stationary occupants	N/A	Weak	Strong	Depends	Good (to verify)
Privacy	Good	Good	Concerning	Depends	Good (no images)
Needs internet	No	No	Often	Yes	No
Setup effort	Low	Low	High	Very high	Moderate
Main risk	Wastes energy	False-offs	Privacy, cost	Complexity	Radar false triggers

Feasibility and plan

Technical

Common modules, plenty of documentation. Main risk is radar tuning, which has scheduled time.

Economic

Prototype ₹2,500–4,000. Deployment payback depends on real waste, so audit first.

Safety

12 V DC loads. Any 230 V work needs rated parts and a qualified electrician; students never touch live mains.

Skills

Basic C/C++, breadboard or soldering, data logging, with guidance.

12-week timeline

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- Phase 0 · Weeks 1–2**
Audit and research: room audit, tariffs, wattages, datasheets, literature → measured Table 1
 - Phase 1 · Week 3**
Design: components, circuit, state logic, test plan → design doc, BOM
 - Phase 2 · Week 4**

Procurement and bench setup → individual sensors working

Phase 3 · Weeks 5–7

Integration: fusion logic, relays, override → working prototype

Phase 4 · Week 8

Tuning: radar gates, timers, placement → tuned parameters

Phase 5 · Weeks 9–10

Experiments: three configurations → raw data

Phase 6 · Week 11

Analysis and documentation → results, graphs, energy

Phase 7 · Week 12

Demo and report → deliverables

Team roles

- Hardware and circuit
- Firmware and fusion logic
- Testing and data analysis
- Research, energy audit and economics
- Enclosure, integration, documentation and demo

Testing and evaluation

Three configurations (PIR-only, radar-only, hybrid) under identical conditions, at least 20 trials per scenario per configuration (to fix with the guide).

Table 9: Test scenarios

Scenario	Purpose
Person enters and walks	Detection and response time
Sits still reading (5–10 min)	Core stationary-occupant test
Typing or writing at desk	Small-motion detection
Group of 3–5	Multi-occupant behaviour
Leaves, room empty	Correct turn-off and delay
Empty room, fan running	False occupancy from fan
Person in corridor outside	Boundary and through-wall leakage
Sunlight or warm object near PIR	PIR false trigger
One sensor disabled	Fault handling

Table 10: Metrics

Metric	Definition	Target
False-vacancy rate	Occupied periods with loads wrongly off	Hybrid well below PIR-only; ≈0 for seated-still
False-occupancy time	Extra ON time in empty room beyond timeout	Report honestly
Response time (ON)	Entry to load ON	Below ~2 s
Turn-OFF accuracy	Actual vs configured delay	Within ±5 s
Energy saved	(Always-ON – actual) / always-ON, from relay logs	Report measured value
Standby power	Power of control node	Compare with 1.5 W

Targets are goals, not predictions. If fan reflections force conservative radar settings, the report will say so.

Expected results (hypotheses)

- Hybrid loses seated-still occupants less often than PIR-only.
- Hybrid may have more extra ON time than PIR-only if radar triggers on fans.
- The prototype shows how the energy estimates become a measurable test.

Applications: classrooms, labs, seminar rooms, offices, libraries, hostel common areas, restrooms, corridors, each with its own timeout.

Risks, safety, privacy, ethics

Risk	Impact	Mitigation
Radar false triggers	Wasted energy	Gate tuning, range limits, placement tests
False vacancy	Annoyance, system disabled	Both-agree off-rule, manual override
Mains hazard	Injury, fire	12 V prototype; qualified electrician for AC
Sensor or controller failure	Lights stuck off or on	Fail-ON default, degraded mode
Assumed numbers differ	Wrong savings claims	Phase 0 audit, scenarios
Standby consumption	Cuts net savings	Count it; low-power design
Privacy	Occupant concerns	No cameras or identity; only presence logged
RF exposure	Student concern	Low-power commercial modules; check datasheet compliance
Data protection	Compliance	DPDP Act 2023: no personal data collected; recheck if identification added

Limitations: one test room will not generalize to all rooms; no AC control; costs and tariffs are estimates.

Future improvements

Daylight sensing (LDR)

AC control via IR remote emulation or thermostat setpoint

Timetable-aware timeouts

Wi-Fi/MQTT energy dashboard

Current sensing of real load

Multi-sensor coverage for large halls

Adaptive timeouts from usage

Certified mains-rated hardware

Conclusion

EcoSwitch tackles a real problem, electricity wasted on empty rooms, with a modest, testable design. PIR brings low cost and speed; mmWave radar brings sensitivity to still occupants, with no cameras. The energy model shows value depends on actual waste, so the project starts with measurement and reports honestly. It is safe at prototype level and realistic for a five-member first-year team.

Request: approval to proceed with Phase 0 (campus audit) and Phase 1 (design).

EcoSwitch project research · PCCOE, Pune · Figures marked as assumptions need verification